

## Task 2

### Multiple Sequence Alignment of Ras-Family Proteins

#### Aim

To perform multiple sequence alignment of five human Ras-family proteins using Clustal Omega, identify conserved regions and functional motifs, and evaluate sequence variation across the family.

#### Introduction

Multiple Sequence Alignment (MSA) aligns three or more biological sequences to identify conserved residues, evolutionary relationships, and critical structural or functional motifs. Ras-family small GTPases operate as molecular switches in intracellular signal transduction pathways, controlling cell growth, differentiation, and survival. Aligning canonical human KRAS against four related Ras-family proteins (HRAS, NRAS, RRAS, and MRAS) highlights highly conserved catalytic domains along with variable regulatory regions. Clustal Omega utilizes progressive alignment and hidden Markov models (HMM) to align these related protein sequences accurately.

#### Objective

- Collect FASTA sequences for five human Ras-family proteins from UniProt.
- Perform multiple sequence alignment using EMBL-EBI Clustal Omega.
- Identify conserved amino acid residues, including the phosphate-binding loop (P-loop) motif.
- Interpret the structural and functional significance of conserved versus variable regions.

| No. | Protein     | Gene | Organism            | Accession | Length |
|-----|-------------|------|---------------------|-----------|--------|
| 1   | GTPase KRas | KRAS | <i>Homo sapiens</i> | P01116    | 189 aa |
| 2   | GTPase HRas | HRAS | <i>Homo sapiens</i> | P01112    | 189 aa |



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## Conserved Motif Analysis

- ## Interpretation

The multiple sequence alignment demonstrates substantial sequence conservation among the five human Ras-family proteins, particularly within the N-terminal GTPase domain (residues 1–120). The strict conservation of the consensus P-loop motif (GAGGVGKS) highlights its indispensable role in binding

the phosphate groups of GDP and GTP. High sequence conservation across Switch I and Switch II loops further confirms a shared mechanism for effector binding and signal transmission. In contrast, sequence divergence at the C-terminus provides functional specificity, allowing different Ras isoforms to localize to distinct subcellular membrane compartments.

## **Conclusion**

Clustal Omega MSA confirmed strong evolutionary conservation across the catalytic core of human Ras-family proteins. The strict preservation of the GTP-binding P-loop motif (GxxxxGKS) and catalytic switch regions supports their shared evolutionary lineage and conserved molecular function as GTPase switches.